

GROUP O

Stakeholder Engagement Record

Meetings, correspondence and findings — 19 August to 3 September 2026

Project title	Multi-Site Smart Soil and Environmental Monitoring System
Document reference	ENG-01
Version	1.0
Date	4 September 2026
Stakeholder and end user	Ms Goitsewang Dikane, Plant Sciences (nGAP), Department of Agriculture
Engagement channel	Ms Mpho Mbele, Departmental Manager: Information Technology
Engagements recorded	One consultation meeting and five items of written correspondence
Prepared by	Group O project delivery team

1. Engagement Log

#	Date	Type	With	Subject	Evidence
1	19 Aug 2026	Consultation meeting	Stakeholder	Scope, architecture and timeline briefing	Recording and transcript
2	20 Aug 2026	Email received	Ms Mpho Mbele	Project requirements document supplied	Email and attachment
3	26 Aug 2026	Email sent	Ms Mpho Mbele	Fifteen follow-up questions issued	Email and attachment
4	27 Aug 2026	Email received	Ms Mpho Mbele	Written responses returned	Email and attachment
5	30 Aug 2026	Email sent	Ms Mpho Mbele	Findings summary and specification issued for confirmation	Email and attachment
6	3 Sep 2026	Email received	Ms Goitsewang Dikane	Findings confirmed and requirements approved	Email

2. Consultation Meeting, 19 August 2026

Subject	IoT soil-monitoring platform — scope, architecture and timeline
Present	The stakeholder, and all eight members of the Group O delivery team. One member spoke on behalf of the group during the recording.
Record	Two audio recordings, transcribed. Full transcript held with the project records.
Reference	MIN-02

2.1 What the meeting established

#	Established at the meeting	Where it is applied
1	Sensor hardware is handled by a separate party. The delivery team does not work on the hardware.	Specification §5.3 Exclusions
2	Readings are transmitted by Wi-Fi and staged in Firebase; the team's role is to bring that data into Supabase.	Specification §8.3
3	Where sensor data is not yet available, the team is to work with representative data in the database.	Specification §8.3, change log C01
4	Three locations, five sensors at each, fifteen in total.	Specification §2.1, AC3 and AC4
5	The database must link locations to sensors, and readings must trace to a named sensor at a named location.	Specification §16
6	Device health is tracked separately. Solar panels and batteries are in use; a failing battery must be flagged.	Specification §10.6
7	The web application is the primary home for visualisation; the	Specification §9

#	Established at the meeting	Where it is applied
	mobile application focuses on device and battery health.	
8	Readings must be interpreted, not merely displayed.	Specification §15 and NFR17
9	Thresholds are to be researched from South African sources; the team may set its own where none exist, provided a basis is stated.	Specification §15 and §15.1
10	Thresholds are crop-specific; no single national threshold applies to all crops.	Specification §15
11	A threshold breach must raise an alert. Push notification preferred, email secondary.	Specification §10.8, change log C09
12	Data is fetched at intervals by a scheduled job, at approximately thirty-minute intervals.	Specification §8.3 and FR75
13	Climate temperature per location, beyond soil temperature, to be incorporated for context.	Open — see section 4
14	Indicative timeline: Phase 1 early September, design end September, delivery end October.	Specification §17.4

2.2 Verbatim extract — scope and data

Stakeholder

“The data is transmitted by Wi-Fi from the sensor to the database, so there's a middleman — you guys don't really have to worry about the hardware parts. At the moment we're going to read data into Firebase, so your role will just be to take that data from Firebase into Supabase.”

Stakeholder

“Even if the data isn't ready, we'll just check the type of data that the sensors are supposed to read. So you just have dummy data on your database for those sensors.”

Stakeholder

“We'll be using three locations. We have fifteen sensors, so for each location we'll put five sensors. In your database you must have locations, and then you need to link locations with the sensors.”

2.3 Verbatim extract — interpretation and thresholds

Stakeholder

“We're not going to leave you with just the raw data — you have to tell us the information and analyse what the data set is saying. So say, for example, pH is 6 — you guys must go and find out what pH 6 means.”

Delivery team

“The thresholds — will we have to determine them ourselves, or will you provide them?”

Stakeholder

“You have to go and find out if they exist. If there are no thresholds, then you can come up with your own, just to have something — I can't have you tweaking something with no basis.”

2.4 Verbatim extract — alerting and scheduled collection

Stakeholder

“Your system must also give alerts. If a threshold is exceeded, then you give out an alert, or maybe send a notification.”

Stakeholder

“From the app. Push notifications are fine, or you can send an email — there's an option for emails. But I think a push notification will do.”

Stakeholder

“Even though we have solar panels, we're also going to use batteries — so if those batteries are going to die at some point, you need to flag it.”

Stakeholder

“There's something called a cron job — how to fetch data at intervals, so it fills up your database automatically, instead of only reading once when something happens on the application.”

3. Written Correspondence

3.1 Requirements document received, 20 August 2026

A written project requirements document was supplied by the stakeholder organisation through Ms Mpho Mbele. It runs to thirty-one numbered sections covering the physical deployment, the soil parameters, user roles, authentication, location and sensor management, data collection and storage, battery and device health monitoring, thresholds, alerts, the mobile and web application requirements, comparison, filtering, reporting, visualisation, backend requirements, database entities, validation, security, reliability, usability, performance, timestamps, testing, the required final demonstration, the integration requirement and the required technical components.

3.2 Follow-up questions issued, 26 August 2026

Fifteen open questions were issued in writing because a meeting could not be arranged during that week. They covered the current process, the people affected, the consequences of delay, user roles and numbers, the structure of a stored reading, battery and device status fields, data access, development priorities, the stakeholder's own measure of success, authorship of the requirements document, and willingness to confirm the findings.

3.3 Written responses received, 27 August 2026

The same document was returned with an answer written beneath each question. Six of those answers changed the specification materially.

Answer	Effect on the project
Soil is sampled once at the start of the planting cycle and sent to a laboratory; there is no continuous monitoring between sampling points	Established the current process recorded in specification §2.2
The main problem is the delay in identifying significant changes across the three locations	Became the problem statement in specification §3
Two user roles are sufficient for this version: Administrator and Standard User	Fixed the access model in specification §7.2, change log C05
For the current prototype, realistic dummy sensor data may be used; Firebase integration is not required at this stage	Changed the architecture. Recorded as change log entries C01 and C02
Dummy data should include battery level, sensor status and last-communication time	Confirmed the device-status entity, change log C03
Priority order if time runs short: login, locations, sensors, current readings, battery and sensor status, historical readings, alerts, dashboard	Became the development sequence in specification §14.2

3.4 Findings summary issued, 30 August 2026

A written summary of the delivery team's understanding was issued together with the completed specification, and the stakeholder was asked to confirm that it was accurate or to correct it.

3.5 Confirmation received, 3 September 2026

Ms Goitsewang Dikane, Plant Sciences (nGAP), Department of Agriculture

“Kindly note that I have gone through the document and have realized that some of the minor corrections can be amended while the process unfolds. The calibration and setting of range measurements can also be amended later on once the platform has been established. We shall revisit the pipeline and amend where necessary, for now I believe the document can be used as it meets the entry requirements as the end user.”

What she said	Where it is recorded
The document meets the entry requirements of the end user and can be used as it stands	Specification §21.1
Minor corrections can be amended while the process unfolds	Specification §21.1, recorded as non-blocking
Calibration and range settings can be amended once the platform is established	Specification §15.2 and change log C14
The pipeline will be revisited and amended where necessary	Specification §15.2

4. Items Raised and Not Yet Closed

Two matters raised in the consultation are recorded here because they are not fully reflected in the specification. Both were raised verbally and are not settled in the written record.

Raised	Position	Status
Climate temperature per location, in addition to soil temperature	The stakeholder undertook to obtain temperature data for each location. This would be an additional external data source beyond the sensors.	Not in the current specification. To be raised before Phase 2 design begins and recorded in the change log with the outcome.
Whether the sensor reports humidity as a measured parameter	The consultation described the sensor as reading temperature, humidity, pH and conductivity alongside nitrogen, phosphorus and potassium. The written requirements document, section 3, defines seven parameters and does not list humidity, and the sample reading fields supplied in the written responses list the same seven.	The specification records seven parameters, following the written requirements document. Recorded as change log entry C04 and to be confirmed with the stakeholder.

5. Attendance

Group members present at or involved in each engagement. All eight members took part in every engagement: the consultation was attended by the group with one member speaking on its behalf, and each item of correspondence was drafted and reviewed by the group before it was sent or acted upon.

Member	19 Aug	20 Aug	26 Aug	27 Aug	30 Aug	3 Sep
Nido Maphosa	P	P	P	P	P	P
Caitlin Freeman	P	P	P	P	P	P
Mbasa Tyam	P	P	P	P	P	P
Tiisetso Masita	P	P	P	P	P	P
Machedi Mamello	P	P	P	P	P	P
Ayatolla Pikwa	P	P	P	P	P	P
Katlontle Seiphetlho	P	P	P	P	P	P
Thobani Mjiyakho	P	P	P	P	P	P

6. Evidence Held

Item	Form	Where held
Consultation meeting recordings	Two audio files, 19 August 2026	Project records
Consultation meeting transcript	Transcribed and formatted	Project records
Project requirements document	PDF received 20 August 2026	Repository, documentation folder
Follow-up questions as issued	Word document sent 26 August 2026	Repository, documentation folder
Stakeholder written responses	Word document received 27 August 2026	Repository, documentation folder
Findings summary as issued	Email and attachment, 30 August 2026	Project records
Stakeholder confirmation	Email received 3 September 2026	Project records

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